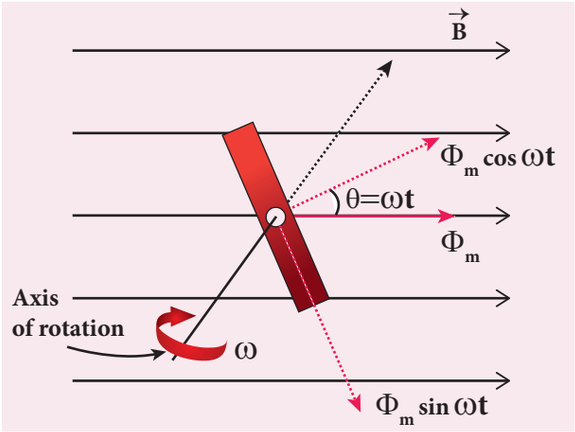


Figure 4.25(b) The coil has rotated through an angle $\theta = \omega t$

At time = 0, the plane of the coil is perpendicular to the field and the flux linked with the coil has its maximum value $\Phi_m = BA$ (where A is the area of the coil).

In a time t seconds, the coil is rotated through an angle θ (= ωt) in anti–clockwise direction. In this position, the flux linked is $\Phi_m \cos \omega t$, a component of Φ_m normal to the plane of the coil (Figure 4.25(b)). The component parallel to the plane ($\Phi_m \sin \omega t$) has no role in electromagnetic induction. Therefore, the flux linkage at this deflected position is

$$N\Phi_B = N\Phi_m \cos \omega t$$

According to Faraday's law, the emf induced at that instant is

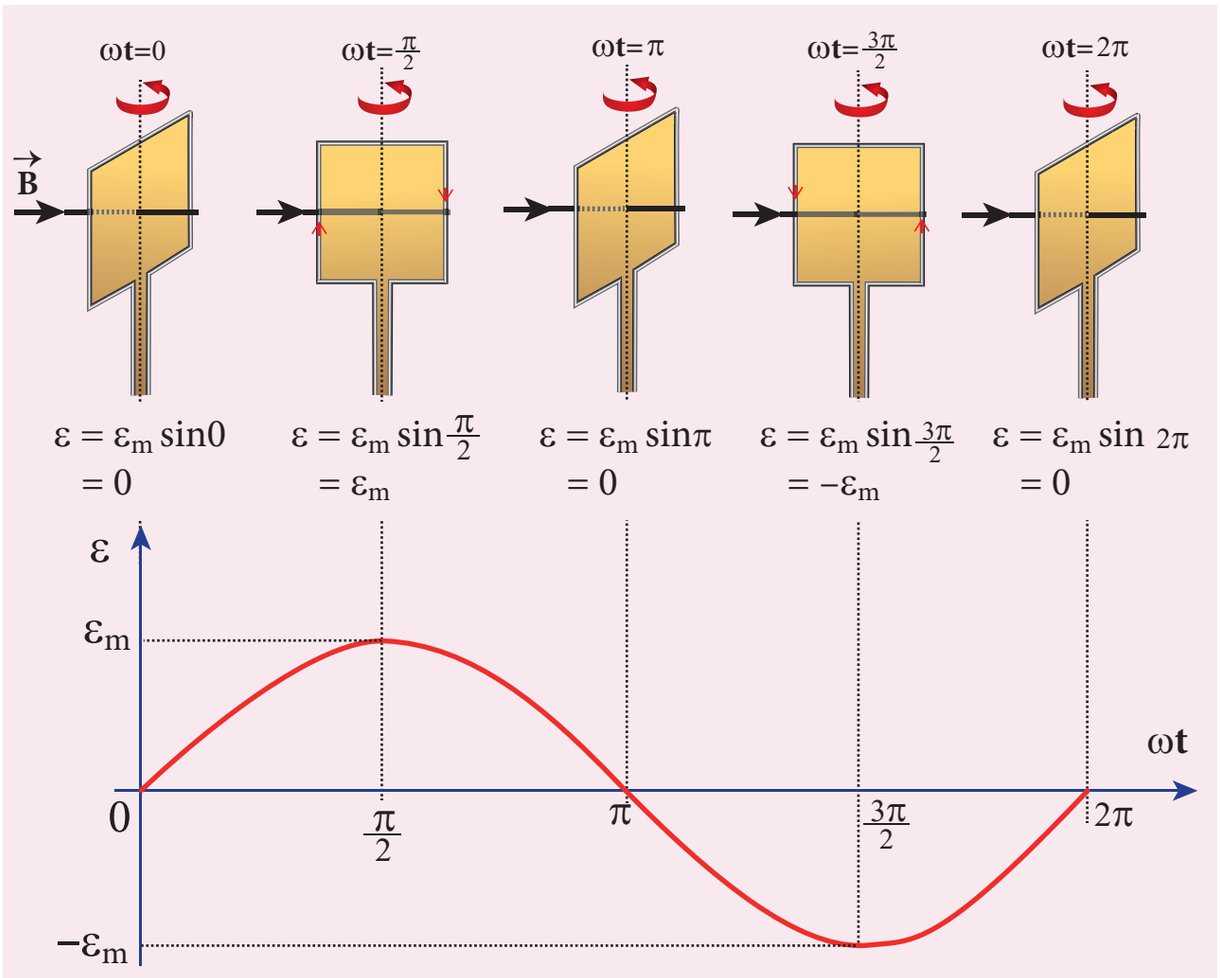


Figure 4.26 Variation of induced emf as a function of ωt

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